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Subject: Mathematics **CBSE CLASS 10 – WEEKLY CHAPTER TEST – 06** **Time:** 45 Minutes

Class: Class X (CBSE) **Chapter 4: Quadratic Equations** **MM:** 20

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

Section A: MCQ & Assertion-Reason Questions

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Q1. The ratio of the sum and product of the roots of the quadratic equation $5x^2 - 6x + 21 = 0$ is: [1]

(A) $5 : 21$ **(B)** $2 : 7$
(C) $21 : 5$ **(D)** $7 : 2$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) $2 : 7$

Comparing $5x^2 - 6x + 21 = 0$ with $ax^2 + bx + c = 0$, we have $a = 5$, $b = -6$, and $c = 21$.

Formula Used: Sum $(\alpha + \beta) = -\dfrac{b}{a}$, Product $(\alpha\beta) = \dfrac{c}{a}$

$$\alpha + \beta = -\frac{-6}{5} = \frac{6}{5}, \qquad \alpha\beta = \frac{21}{5} \implies \frac{\alpha+\beta}{\alpha\beta} = \frac{6/5}{21/5} = \frac{6}{21} = \frac{2}{7} \implies \boxed{2 : 7}$$

Therefore, option **(B)** is correct. **[1 mark]**

Q2. If one root of the quadratic equation $x^2 - 4x + 3 = 0$ is 1, then the other root is: [1]

(A) 3 **(B)** -3
(C) 2 **(D)** 4

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 3

Let the roots of $x^2 - 4x + 3 = 0$ $(a = 1, b = -4, c = 3)$ be $\alpha = 1$ and β.

Formula Used: $\alpha + \beta = -\dfrac{b}{a}$ and $\alpha\beta = \dfrac{c}{a}$

$$1 + \beta = -\frac{-4}{1} = 4 \implies \beta = 4 - 1 = \boxed{3} \quad \left(\text{also check: } \beta = \frac{c}{a \cdot 1} = \frac{3}{1} = 3\right)$$

Therefore, option **(A)** is correct. **[1 mark]**

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Q3. If the roots of $px^2 + qx + 2 = 0$ are reciprocal of each other, then: [1]

(A) $p = 0$ **(B)** $p = -2$

(C) $p = 2$ **(D)** $q = 2$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) $p = 2$

Let the roots of $px^2 + qx + 2 = 0$ $(p \neq 0)$ be α and $\frac{1}{\alpha}$.

Formula Used: Product of roots $= \frac{c}{a} \implies \alpha \cdot \frac{1}{\alpha} = 1$

Here $a = p$ and $c = 2$:

$$\frac{2}{p} = 1 \implies \boxed{p = 2}$$

Hence, option **(C)** is correct. **[1 mark]**

Q4. If $x = 1$ is a common root of the equations $ax^2 + ax + 3 = 0$ and $x^2 + x + b = 0$, then $ab =$: [1]

(A) 3 **(B)** 3.5

(C) 6 **(D)** -3

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 3

Since $x = 1$ satisfies both quadratic equations:

Formula Used: Substitute $x = 1$ into each equation to solve for a and b

- From $ax^2 + ax + 3 = 0$: $a(1)^2 + a(1) + 3 = 0 \implies 2a + 3 = 0 \implies a = -\frac{3}{2}$.
- From $x^2 + x + b = 0$: $(1)^2 + (1) + b = 0 \implies 2 + b = 0 \implies b = -2$.

Therefore, $ab = \left(-\frac{3}{2}\right)(-2) = \boxed{3}$. Hence, option **(A)** is correct. **[1 mark]**

Q5. Assertion-Reason: [1]

Assertion (A): The equation $(x+1)^2 - x^2 = 0$ has degree 1 and has only one real root.

Reason (R): On expanding $(x+1)^2 - x^2 = 0$, the x^2 term cancels leaving the linear equation $2x+1 = 0$.

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true, but R is not the correct explanation of A.

(C) A is true, R is false.

(D) A is false, R is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

Expanding $(x+1)^2 - x^2 = 0$ using $(a+b)^2 = a^2 + 2ab + b^2$:

Formula Used: $(x+1)^2 - x^2 = (x^2 + 2x + 1) - x^2 = 2x + 1 = 0$

The x^2 terms cancel out completely, reducing the equation to a linear equation $2x + 1 = 0$ of degree 1, which has a single real root $x = -\frac{1}{2}$.

Thus Assertion (A) is **true**, Reason (R) is **true**, and Reason (R) correctly explains Assertion (A). **[1 mark]**

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Q6. Find the nature of the roots of $3x^2 - 4\sqrt{3}x + 4 = 0$. If they are real, find them. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Comparing $3x^2 - 4\sqrt{3}x + 4 = 0$ with $ax^2 + bx + c = 0$, we have $a = 3$, $b = -4\sqrt{3}$, and $c = 4$.

Formula Used: $D = b^2 - 4ac, \qquad x = \dfrac{-b \pm \sqrt{D}}{2a}$

Step 1: Compute discriminant D:

$$D = (-4\sqrt{3})^2 - 4(3)(4) = 16(3) - 48 = 48 - 48 = 0 \quad \textbf{[1 mark]}$$

Since $D = 0$, the equation has **two real and equal roots**.

Step 2: Using the quadratic formula with $D = 0$:

$$x = \frac{-(-4\sqrt{3}) \pm \sqrt{0}}{2(3)} = \frac{4\sqrt{3}}{6} = \frac{2\sqrt{3}}{3} = \frac{2}{\sqrt{3}}$$

Hence, the equation has $\boxed{\text{real and equal roots: } x = \frac{2\sqrt{3}}{3}, \frac{2\sqrt{3}}{3}}$. **[1 mark]**

Q7. An express train takes 1 hour less than a passenger train to travel 132 km between Mysore and Bangalore. If the average speed of the express train is 11 km/h more than that of the passenger train, find the average speeds of both trains. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Let the average speed of the passenger train be x km/h ($x > 0$).

Then the average speed of the express train $= (x + 11)$ km/h. Distance $= 132$ km.

Formula Used: $\text{Time} = \dfrac{\text{Distance}}{\text{Speed}}, \qquad t_{\text{passenger}} - t_{\text{express}} = 1 \text{ h}$

Setting up the equation:

$$\frac{132}{x} - \frac{132}{x+11} = 1$$

$$132\left[\frac{(x+11) - x}{x(x+11)}\right] = 1 \implies \frac{132 \times 11}{x^2 + 11x} = 1 \implies x^2 + 11x - 1452 = 0 \quad \textbf{[1 mark]}$$

Splitting the middle term ($44 \times (-33) = -1452$ and $44 - 33 = 11$):

$$x^2 + 44x - 33x - 1452 = 0 \implies x(x+44) - 33(x+44) = 0 \implies (x+44)(x-33) = 0 \quad \textbf{[1 mark]}$$

Thus $x = -44$ or $x = 33$. Since speed cannot be negative ($x > 0$), we reject $x = -44$.

Therefore, passenger train speed $= 33$ km/h, express train speed $= 33 + 11 = 44$ km/h. (Check: $\frac{132}{33} - \frac{132}{44} = 4 - 3 = 1$ h.)

Hence, $\boxed{\text{Passenger train} = 33 \text{ km/h, Express train} = 44 \text{ km/h}}$. **[1 mark]**

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Q8. Solve for x: $\frac{1}{a+b+x} = \frac{1}{a} + \frac{1}{b} + \frac{1}{x}$, $\quad a, b, x \neq 0,\ x \neq -(a+b)$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Transposing $\frac{1}{x}$ to the LHS so that terms involving x are grouped together:

Formula Used: $\frac{1}{a+b+x} - \frac{1}{x} = \frac{1}{a} + \frac{1}{b}$

Taking LCM on both sides separately:

$$\frac{x-(a+b+x)}{x(a+b+x)} = \frac{b+a}{ab}$$

$$\frac{-(a+b)}{x(a+b+x)} = \frac{a+b}{ab} \quad \textbf{[1 mark]}$$

Bringing everything to one side and factoring out $(a+b)$ (assuming $a+b \neq 0$; if $a+b=0$, equation reduces to $1/x = 1/x$ trivially):

$$(a+b)\left[\frac{1}{ab} + \frac{1}{x(a+b+x)}\right] = 0$$

$$x(a+b+x) + ab = 0 \implies x^2 + (a+b)x + ab = 0 \quad \textbf{[1 mark]}$$

Factorising by grouping:

$$x^2 + ax + bx + ab = 0 \implies x(x+a) + b(x+a) = 0 \implies (x+a)(x+b) = 0$$

Therefore, $x = -a$ or $x = -b$ (note $x \neq -(a+b)$ since $a, b \neq 0$).

Hence, the solutions are $\boxed{x = -a \text{ and } x = -b}$. **[1 mark]**

Q9. In a flight of 600 km, an aircraft was slowed down due to bad weather. Its average speed was reduced by 200 km/h and the time of flight increased by 30 minutes. Find the original duration of the flight. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Let the original average speed of the aircraft be v km/h ($v > 200$).

Then reduced speed = $(v - 200)$ km/h, distance = 600 km, and increase in time = 30 min = $\frac{1}{2}$ hour.

Formula Used: $\text{Time} = \frac{\text{Distance}}{\text{Speed}}, \qquad t_{\text{delayed}} - t_{\text{original}} = \frac{1}{2}$ h

Setting up the equation:

$$\frac{600}{v-200} - \frac{600}{v} = \frac{1}{2}$$

$$600\left[\frac{v-(v-200)}{v(v-200)}\right] = \frac{1}{2} \implies \frac{120000}{v^2 - 200v} = \frac{1}{2} \implies v^2 - 200v - 240000 = 0 \quad \textbf{[1 mark]}$$

Factorising ($-600 \times 400 = -240000$ and $-600 + 400 = -200$):

$$(v-600)(v+400) = 0 \implies v = 600 \text{ or } v = -400 \quad \textbf{[1 mark]}$$

Since speed cannot be negative, we reject $v = -400$. Thus original speed $v = 600$ km/h.

Therefore, the original duration of the flight $= \frac{\text{Distance}}{\text{Original speed}} = \frac{600}{600} = \boxed{1 \text{ hour}}$. **[1 mark]**

Q10. Case Study: Civil Aviation & Weather Delay Dynamics

A commercial passenger airliner operates on a scheduled domestic sector covering an aerial distance of 1500 km. On a monsoon day, due to adverse weather and runway congestion, the aircraft departed 30 minutes (0.5 hour) later than its scheduled departure time. To reach the destination on time, the pilot increased the cruising speed by 100 km/h over the scheduled speed v km/h. [4]

(i) Write expressions for the scheduled flight time and actual flight time in terms of v. **[1]**

(ii) Formulate the quadratic equation in v in standard form modeling the time constraint. **[1]**

(iii) Determine the scheduled cruising speed of the aircraft (v). **OR** Find the scheduled duration of the flight. **[2]**

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

We know the standard time–distance–speed relation:

Formula Used: $\text{Time} = \dfrac{\text{Distance}}{\text{Speed}}$

(i) Flight Time Expressions: Distance = 1500 km, scheduled speed = v km/h ($v > 0$), increased speed = $(v + 100)$ km/h.

$$\boxed{t_{\text{scheduled}} = \frac{1500}{v} \text{ hours}} \quad \text{and} \quad \boxed{t_{\text{actual}} = \frac{1500}{v+100} \text{ hours}} \quad \textbf{[1 mark]}$$

(ii) Quadratic Equation in Standard Form: To cover the 30 min $= \dfrac{1}{2}$ hour departure delay and still arrive on time:

$$\frac{1500}{v} - \frac{1500}{v+100} = \frac{1}{2}$$

$$1500\left[\frac{(v+100) - v}{v(v+100)}\right] = \frac{1}{2} \implies \frac{150000}{v^2 + 100v} = \frac{1}{2} \implies \boxed{v^2 + 100v - 300000 = 0} \quad \textbf{[1 mark]}$$

(iii) First Part (Scheduled Speed): Splitting the middle term ($600 \times (-500) = -300000$ and $600 - 500 = 100$):

$$v^2 + 600v - 500v - 300000 = 0 \implies (v+600)(v-500) = 0 \implies v = -600 \text{ or } v = 500$$

Since aircraft speed cannot be negative ($v > 0$), we reject $v = -600$.
Hence, the scheduled cruising speed is $\boxed{500 \text{ km/h}}$. **[2 marks]**

OR (Alternative Part — Scheduled Duration): Using $v = 500$ km/h from part (iii):

$$t_{\text{scheduled}} = \frac{1500}{500} = 3 \text{ hours} \qquad \left(\text{Check: } t_{\text{actual}} = \frac{1500}{600} = 2.5 \text{ hours, } 3 - 2.5 = 0.5 \text{ h}\right)$$

Hence, the scheduled duration of the flight is $\boxed{3 \text{ hours}}$. **[2 marks]**

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